Sentiment Classification of Steam Game Reviews: Classical Models vs. Fine-Tuned BERT

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Abstract

We present a study on binary sentiment classification of Steam game reviews, comparing classical machine learning models against a fine-tuned BERT transformer. We compiled a corpus of 555,498 English-language reviews across 37 games by scraping the Steam public API. The dataset has a large class imbalance (87.9% positive) and extreme length variability (median 9 words). Among classical approaches, a LinearSVC with TF-IDF features achieves a macro F1 of 0.9385. Fine-tuning `bert-base-uncased` yields a macro F1 of 0.9909, with the biggest gains on the minority (Not Recommended) class (F1: 0.9839 vs. 0.8927). We also plan to test generalizability on an independent 106K-review Kaggle dataset.

1 Introduction

Sentiment analysis of user-generated reviews has real commercial value. Game developers and publishers use aggregated opinion signals to guide post-launch patches, marketing decisions, and platform choices. Beyond commercial interest, online game reviews are an interesting test case for natural language processing: they are informal, platform-specific, and often ironic, which sets them apart from the movie review datasets that have driven most prior sentiment research.

Steam, Valve's PC gaming platform, is well suited to large-scale sentiment study. Its public `appreviews` API gives open access to hundreds of millions of reviews, each tagged with a binary thumbs up or thumbs down chosen by the reviewer. Steam reviews differ from the standard IMDB benchmark (Maas et al., 2011) in several ways: they tend to be very short (median 9 words), lean heavily positive, and are full of gaming-specific slang, memes, and irony.

These features lead to three research questions. **RQ1**: Does the performance gap between classical bag-of-words models and BERT-style transformers hold up on short, imbalanced Steam reviews? **RQ2**: How does severe class imbalance (87.9% positive) affect Not Recommended classification across model types? **RQ3**: How does extreme length variability (median 9 words, tail to 2,000) affect the two approaches differently?

This paper makes the following contributions: (1) a large, genre-diverse dataset of 555K Steam reviews with deliberate inclusion of mixed-reception games to reduce positivity bias; (2) a comparison of Naive Bayes, Logistic Regression, SVM, and fine-tuned BERT on this dataset; and (3) a per-game accuracy analysis showing how the level of review agreement affects how hard classification is.

2 Dataset

2.1 Collection

We collected reviews from the Steam public `appreviews` endpoint, which requires no authentication and supports cursor-based pagination to retrieve full corpora. We targeted 37 games across AAA, indie, strategy, and horror genres. To reduce the platform's tendency toward positive reviews, we included games with mixed or negative receptions, such as Redfall, Skull and Bones, and Saints Row (2022), alongside more well-regarded titles. All reviews were filtered to English only. A second dataset of around 106K reviews (2.2 GB) from Kaggle will be used in later experiments to test how well the trained models generalize.

2.2 Labels and Statistics

Each review is labeled by the boolean field `voted_up`: `True` means Recommended and `False` means Not Recommended. After removing null and empty review texts, the corpus has 555,498 reviews. The class split is very uneven: 87.9% Recommended and 12.1% Not Recommended. Review length is also very skewed: the median is 9 words, about 200,000 reviews have fewer than 5 words, and the longest review has 1,995 words.

2.3 Preprocessing and Splits

Beyond removing null and empty texts, we did no additional preprocessing such as lowercasing or stemming. We left those choices to each model's vectorizer so that the comparison would be fair. The dataset was split 80/10/10 using stratified sampling to keep the class distribution the same across splits, giving 443,983 training, 55,498 development, and 55,498 test examples. All results are reported on the held-out test set.

3 Methods

3.1 TF-IDF Feature Extraction

All classical models use a shared feature representation from scikit-learn's `TfidfVectorizer`, set up with 50,000 features, unigram and bigram tokenization, sublinear TF scaling ($\log(1+tf)$), and a minimum document frequency of 5. The vectorizer is fit on the training set only and then applied to the development and test sets.

3.2 Naive Bayes

As a fast and simple baseline, we train a Multinomial Naive Bayes classifier with Laplace smoothing ($\alpha = 1.0$) on the TF-IDF features.

3.3 Logistic Regression

We train an L2-regularized logistic regression model ($C = 1.0$) using the `lbfgs` solver with up to 1,000 iterations. To handle class imbalance, we use `class_weight='balanced'`, which weights each class inversely by how often it appears in training.

3.4 SVM

We use a `LinearSVC` ($C = 1.0$, `class_weight='balanced'`). Linear kernels work well in high-dimensional, sparse TF-IDF feature spaces and have consistently performed well on binary sentiment classification (Pang et al., 2002).

3.5 BERT

We fine-tune `bert-base-uncased` (Devlin et al., 2019) using the HuggingFace Transformers library with a maximum sequence length of 256 tokens. We use the AdamW optimizer (lr $= 2 \times 10^{-5}$, warmup_ratio $= 0.1$, weight_decay $= 0.01$) for 3 epochs with a batch size of 16 and fp16 mixed precision on an NVIDIA RTX 4080 Super. To handle class imbalance, we use weighted cross-entropy loss with class weights of 4.12 for Not Recommended and 0.57 for Recommended, set inversely by class frequency. The best checkpoint is picked by macro F1 on the development set.

Model	**Acc.**	**Mac. F1**	**F1$_0$**	**F1$_1$**
Naive Bayes	0.9548	0.8784	0.7819	0.9748
Logistic Regression	0.9592	0.9123	0.8481	0.9765
SVM	0.9724	0.9385	0.8927	0.9842
BERT	0.9961	0.9909	0.9839	0.9978

Table 1: Test-set results. $F1_0$ = Not Recommended; $F1_1$ = Recommended.

4 Results

Table 1 shows accuracy, macro F1, and per-class F1 for all four models on the held-out test set. Among classical models, SVM gets the highest macro F1 of 0.9385, ahead of Logistic Regression (0.9123) and Naive Bayes (0.8784). All classical models show a clear gap between their Not Recommended F1 ($F1_0$) and Recommended F1 ($F1_1$), which confirms that class imbalance hurts minority-class performance even when class weighting is applied.

Fine-tuned BERT clearly outperforms the best classical model, reaching a macro F1 of 0.9909 (+5.2 points over SVM) and an accuracy of 0.9961. The gain is biggest on the minority class: BERT's Not Recommended F1 of 0.9839 beats SVM's 0.8927 by 9.1 points. This suggests that contextual representations pick up on signals that sparse bag-of-words features miss, and that the weighted loss is especially useful for recovering minority-class examples under heavy imbalance.

5 Discussion

SVM vs. Logistic Regression. Even though both models use the same TF-IDF features, SVM beats Logistic Regression by 2.6 macro F1 points. This matches the finding from Pang et al. (2002), who showed that margin-based classifiers tend to generalize better on high-dimensional sparse text data. SVM's focus on the decision margin likely gives it better built-in regularization than the L2 penalty used in logistic regression.

Per-game accuracy variance. Looking at per-game results shows a wide range: Logistic Regression accuracy goes from 0.852 on Dead by Daylight all the way to 1.000 on Dave the Diver, Disco Elysium, Europa Universalis IV, and Age of Empires IV. Games with perfect accuracy probably have reviews that almost all point in the same direction, making classification easy. Dead by Daylight, a live-service game with a history of balance disputes and a divided community, produces more mixed reviews that are harder to sort correctly. This spread is why we use macro F1 rather than accuracy as our main metric.

Short reviews. About 200,000 reviews have fewer than five words and give models very little to work with. For TF-IDF models, a one-word review provides almost no useful features. BERT handles these short sequences without truncation, but self-attention over just one or two tokens also has little to work with. Very short reviews are likely a big source of errors across all models.

Sarcasm and label noise. Steam has a culture of ironic reviews where users write negative text but still click Recommended as a joke, or the other way around. Because `voted_up` is chosen by the reviewer and is not inferred from the text, these cases create label noise that no model can fix. This puts a cap on how well any text-based classifier can do.

BERT truncation at 256 tokens. Reviews longer than 256 tokens get cut off, which may cause errors when the most important opinion comes near the end. Given the short median length, this should affect only a small share of reviews. Future work could use hierarchical or sliding-window BERT to handle longer inputs.

6 Related Work

Pang et al. (2002) showed that SVM with bag-of-words features is a strong baseline for binary sentiment classification on movie reviews, outperforming Naive Bayes and Logistic Regression. Our results confirm this same ordering on Steam data. Maas et al. (2011) introduced the IMDB benchmark and found that bag-of-words models struggle with longer, nuanced reviews. Steam reviews are much shorter and more skewed, which brings a different set of challenges. Peters et al. (2018) showed that context-aware word embeddings (ELMo) transfer well to classification tasks, pointing the way toward pre-trained language models. Devlin et al. (2019) then introduced BERT, which reached state-of-the-art results on many sentence- and document-level sentiment tasks. Despite Steam's scale, Steam-specific NLP research is sparse, and no prior work has done a large-scale comparison of classical and transformer-based models on Steam reviews.

7 Conclusion

We compared three classical machine learning models and fine-tuned BERT on a self-scraped corpus of 555,498 Steam reviews across 37 games. SVM with TF-IDF features is a solid classical baseline (macro F1 0.9385), but fine-tuned BERT clearly outperforms it with a macro F1 of 0.9909, a gain of 5.2 points. The biggest improvement is on the minority Not Recommended class (+9.1 F1 points), showing that contextual representations are especially helpful when class imbalance is severe. BERT's extra compute cost is worth it when catching the minority class matters, such as identifying games at risk of a community backlash. Future work includes treating very short reviews as a separate case, adding sarcasm detection as a preprocessing step, and testing on the Kaggle holdout set to measure cross-corpus generalizability.

Limitations

Our study has a few limitations. BERT's 256-token limit cuts off longer reviews, which may cause errors when key opinions appear late in the text. Sarcasm and ironic reviews create label noise that sets a ceiling on how well any model can perform. Our dataset covers only 37 games and only English reviews, so results may not hold across all Steam genres or languages. Finally, `voted_up` reflects what the reviewer chose to click, not necessarily what they wrote, which creates a gap between the label and the text signal.

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